

Same-Frame Pixel-Level A/B Display

the picture itself becomes the measurement instrument

A **Zynq-7020** fabric receives 512×300 RGB565 video over GbE, SD card or its own generated test pattern, applies geometry and effects entirely in the PL, and drives 1024×600 HDMI in which **every column left of an adjustable seam is the unprocessed frame and every column right of it is the processed one** — one read address, one clock cycle, one scanline. No capture card, no screen recording, no second panel: misalignment, dropped columns and out-of-range fill become visible *on the panel*.

Device **xc7z020c1g484-2** Tools **Vivado / Vitis 2025.2.1** RTL **80 files · 11,768 lines** Firmware **2,544 lines bare-metal C** Testbenches **82** Full-screen pixel criteria **141**

INGRESS
RGMII → GMII
IDDR + IDELAY, frame reassembly, port filter, ARP & ICMP answered in RTL

→

STORAGE
DDR ping-pong
3 banks, BRAM row rings, single-owner arbiter ETH | SD | TEST

→

GEOMETRY
Zoom · Rotate · Bilinear
8 fixed-point ratios 0.25–2.00×, angle-driven auto-fit

→

EFFECTS
6-stage chain
9-bit control word, fixed 15-cycle latency at any combination

→

DISPLAY
A/B seam → OSD → TMDS
1024×600 @ 59.5 Hz from a 50 MHz pixel clock, ×2 serialise

01 WHAT IS NEW

One frame, two images display = instrument

The seam mixes original and processed pixels **per column, per clock**; its position is a runtime parameter (0–100 %) and can be pinned in image space so it survives zoom and rotation. That turns a demo trick into a measuring device: geometry errors appear as a step, a gap or a black wedge *across the seam*.

Kept criterion: after `split <n>` the seam marker must not move — a parameter change without a datapath change fails.

One coordinate operator for zoom, rotation and interpolation fixed point

`zfit` solves the fit factor in 3 DSP48 blocks, `zmap` maps every pixel through 8 DSP48 blocks, rotation runs inverse-mapped with two linear interpolations. Eight ratios from 0.25× to 2.00× plus auto-breath and auto-fit, so a rotated frame **always stays inside the panel** — checked by a two-sided bench test, not by an eye.

Ratio, source and fit state read back over AXI GPIO: the on-screen cell and the register value are the same number.

A datapath whose latency is a constant 15 cycles, always

Six pipeline stages, nine independently switchable algorithms, each window stage keeping its own column tag — so total latency is **15 cycles and content offset –4 rows for every combination**. The fabric exposes one number, `u_pipe.LATENCY`; the read scheduler never re-derives it, and any drift fails a bench test.

That invariant is what keeps the A/B seam honest while effects are switched live.

02 MEASURED, WITH ITS CONDITION

ITEM	VALUE	CONDITION
Panel	1024×600 @ 59.5 Hz	50 MHz pixel clock, H 1344 / V 625
Ingress, zero loss	0 dropped · 0 corrupt	600-frame run, 512×300 @ 30 fps, 15 MB/s
Long run	29.99 fps · 0 repeat	300 s, 9,000 frames, counted in hardware
Ingress head-room	≥ 116.7 fps	unthrottled, 960 frames, no lost word; overload point not reached — a bound, not a maximum
SD playback	29.8–30.0 fps	pre-converted RGB565, no decode on board
PL frame interval	33.34 ms	received frames only, min 20 / max 51 ms — excludes PC encode, cable and switch queueing
Timing closure	WNS +0.739 · WHS +0.052 ns	post-route; 0 of 51,135 setup endpoints failing; 100 / 125 MHz domains
Cold-boot standalone	QSPI → FSBL → bit → app	power cycle, no host: <i>Boot mode is QSPI → FPGA Done! → SUCCESSFUL_HANDOFF</i> , SD playback starts itself
Junction temperature	61.2–61.7 °C	on-die XADC, six readings; tool estimate 52.6 °C and 2.213 W dynamic (confidence Low) are separate quantities

LUT		14,154 of 53,200 · 26.61 %
Registers		8,188 of 106,400 · 7.70 %
BRAM		95.5 of 140 tiles · 68.21 %
DSP48		19 of 220 · 8.64 %

24 items / 1 red

pre-release gate: the single red is declared, not deleted — frame-head rows read the previous frame's geometry (kept on the record)

6 cuts rejected

tried and turned down after measurement: per-beat checksum spread · compile-time constant hoisting · forced replication of the 239-fanout net · Pblock · BRAM-for-setup · ±0.5 ns input window — readings published, not hidden

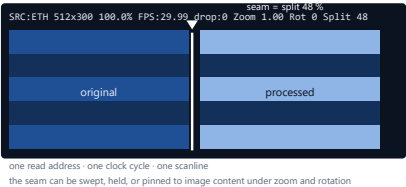
03 VERIFICATION & REPRODUCIBILITY

Three layers that do not substitute for each other: **82 unit testbenches**; a **full-screen bench** walking the whole design into 141 per-pixel criteria with RTL and bench md5 in the report header; and a **board layer only a person can judge** — 36 eye-check rows, each giving command, expected image and what a "no" would mean.

Every criterion must be able to turn red: passing on an empty set counts as failure, and each fix carries a mutation control that reddens exactly the criterion it claims. One command rebuilds every report; one command exports the submission and refuses to write it if a dead link, a stale name or an absolute path remains.

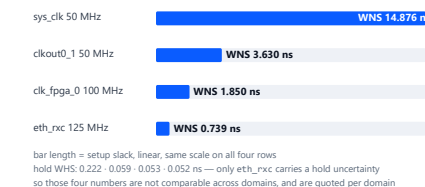
SKILL PACK 49 practice entries decoupled from this topic — scope, usage, **failure conditions**, verified reuse, including the negative results.

05 WHAT THE PANEL SHOWS



Schematic of the on-screen state, drawn from the OSD cells and the seam parameter — not a photograph; no frame-grab instrument was used.

06 PER-DOMAIN SLACK, POST-ROUTE



0 of 51,135 setup endpoints failing. The 125 MHz domain is the tight one: its worst family is a row-coverage enable broadcast, not an adder — attribution recorded with the ladder of 12 tiers.

07 REPRODUCE & LESSONS

```
vivado -mode batch -source build/build.tcl
project → synth → implement → bitstream, about 20 min
bash build/gates.sh

24 pre-release items; the last line is the verdict
bash build/sim/run_one.sh tb_video_pipeline_top
full-screen bench, 141 per-pixel criteria, about 2 h
bash build/sim/mut_control.sh <tb> <criterion>
puts the bug back in: that criterion must turn red, and only it
bash build/make_submission.sh
export + four self-checks; refuses to write a package with a dead link
```

Three things this project actually learned. ① A number without its condition is not a result — every metric row names the file it came from, and absolute WNS deltas are deliberately not counted as gain. ② The check needs a test too: a criterion that cannot turn red, or that passes on an empty set, is treated as a failure of the tooling, not of the design. ③ Negative results are cheap to keep and expensive to lose — six timing cuts are published as rejected with their readings, which is what makes the accepted ones credible.